Math 271: Mathematical Methods
Class Test 6 (05 October 2026) — Model Solutions

The four parts are proved in order, and each later part builds on the earlier ones: part (a) restricts a concave function to a line, part (b) proves the tangent-line inequality in one variable, part (c) reduces the n-variable tangent-plane inequality to (b) by means of (a), and part (d) combines (c) with Taylor's theorem.

Part (a)

Claim. Let $f : \mathbb{R}^n \to \mathbb{R}$ be concave, fix $x, y \in \mathbb{R}^n$, and define $\varphi : [0,1] \to \mathbb{R}$ by $\varphi(t) = f\big(tx + (1-t)y\big)$. Then φ is concave on $[0,1]$.

Proof. First, $[0,1]$ is a convex subset of $\mathbb{R}$: if $s, t \in [0,1]$ and $0 \le \lambda \le 1$ then $\lambda s + (1-\lambda)t$ is a weighted average of two numbers in $[0,1]$, so it lies in $[0,1]$. Hence it makes sense to ask whether φ is concave on $[0,1]$.

For $t \in [0,1]$ write $z(t) = tx + (1-t)y \in \mathbb{R}^n$, so that $\varphi(t) = f(z(t))$.

Now take $s, t \in [0,1]$ and $\lambda \in [0,1]$, and put $r = \lambda s + (1-\lambda)t \in [0,1]$. The key observation is that the map $t \mapsto z(t)$ is affine, so it sends weighted averages to weighted averages:

$$\begin{aligned} \lambda z(s) + (1-\lambda) z(t) &= \lambda\big(sx + (1-s)y\big) + (1-\lambda)\big(tx + (1-t)y\big) \\ &= \big(\lambda s + (1-\lambda)t\big)x + \big(\lambda(1-s) + (1-\lambda)(1-t)\big)y \\ &= \big(\lambda s + (1-\lambda)t\big)x + \big(1 - \lambda s - (1-\lambda)t\big)y \;=\; r\,x + (1-r)\,y \;=\; z(r). \end{aligned}$$

Therefore, using the concavity of f with the points $z(s), z(t) \in \mathbb{R}^n$ and the weight $\lambda \in [0,1]$,

$$\varphi\big(\lambda s + (1-\lambda)t\big) = f\big(z(r)\big) = f\big(\lambda z(s) + (1-\lambda) z(t)\big) \;\ge\; \lambda f\big(z(s)\big) + (1-\lambda) f\big(z(t)\big) = \lambda\varphi(s) + (1-\lambda)\varphi(t).$$

Since $s, t \in [0,1]$ and $\lambda \in [0,1]$ were arbitrary, φ is concave on $[0,1]$. This holds for every choice of $x, y \in \mathbb{R}^n$. □

Remark. The argument uses nothing about the domain $[0,1]$ (for φ) or the domain $\mathbb{R}^n$ (for f) apart from convexity. For later use we record the general version, which has exactly the same proof.

Lemma 1 (Restriction to a line)**.** *Let $A \subseteq \mathbb{R}^n$ be convex, $F : A \to \mathbb{R}$ concave, $x \in A$ and $v \in \mathbb{R}^n$. Let $J = \{t \in \mathbb{R} : x + tv \in A\}$ and $\psi(t) = F(x + tv)$ for $t \in J$. Then J is a convex subset of $\mathbb{R}$ (that is, an interval) containing 0, and ψ is concave on J. If moreover A is open, then J is open.*

Proof. Clearly $0 \in J$ because $x \in A$. Let $w(t) = x + tv$. For $s, t \in \mathbb{R}$ and $\lambda \in [0, 1]$,

$$\lambda w(s) + (1-\lambda)w(t) = x + \big(\lambda s + (1-\lambda)t\big)v = w\big(\lambda s + (1-\lambda)t\big).$$

If $s, t \in J$ then $w(s), w(t) \in A$, so the left-hand side is in A by convexity of A. Hence $\lambda s + (1-\lambda)t \in J$, so J is convex. Using the same identity and the concavity of F,

$$\psi\big(\lambda s + (1-\lambda)t\big) = F\big(\lambda w(s) + (1-\lambda)w(t)\big) \geq \lambda F(w(s)) + (1-\lambda)F(w(t)) = \lambda\psi(s) + (1-\lambda)\psi(t).$$

Finally, suppose A is open; we show directly from the definition that J is open. If $v = 0$ then $x + tv = x \in A$ for every t, so $J = \mathbb{R}$, which is open. Let $v \neq 0$ and $t_0 \in J$. Then $x + t_0 v \in A$, and since A is open there is $r > 0$ such that every $z \in \mathbb{R}^n$ with $\|z - (x + t_0 v)\| < r$ lies in A. Put $\varepsilon = r/\|v\| > 0$. If $|t - t_0| < \varepsilon$, then

$$\big\|(x + tv) - (x + t_0 v)\big\| = |t - t_0|\,\|v\| < \varepsilon\|v\| = r,$$

so $x + tv \in A$, i.e. $t \in J$. Thus $(t_0 - \varepsilon, t_0 + \varepsilon) \subseteq J$ for every $t_0 \in J$, which means J is open. (Being convex and open, J is an interval that contains none of its end-points, i.e. an open interval, possibly unbounded.) □

Part (b)

Claim. Let $I \subseteq \mathbb{R}$ be an open interval and g a C^1 function on I. If g is concave on I, then

$$g(y) \leq g(x) + g'(x)(y - x) \qquad \text{for all } x, y \in I.$$

Proof. Fix $x, y \in I$. If $y = x$, both sides equal $g(x)$ and there is nothing to prove, so assume $y \neq x$.

Step 1 (concavity along the segment). For $t \in (0, 1]$ put

$$x_t = x + t(y - x) = t\,y + (1 - t)\,x.$$

Since I is an interval (hence convex) and $x, y \in I$, we have $x_t \in I$. Concavity of g with weight $t \in [0, 1]$ gives

$$g(x_t) = g\big(ty + (1-t)x\big) \;\geq\; t\,g(y) + (1 - t)\,g(x),$$

that is,

$$g(x_t) - g(x) \;\geq\; t\big(g(y) - g(x)\big).$$

Dividing by $t > 0$ (which does not change the direction of the inequality),

$$\frac{g\big(x + t(y - x)\big) - g(x)}{t} \;\geq\; g(y) - g(x) \qquad \text{for all } t \in (0, 1]. \tag{1}$$

Step 2 (letting $t \to 0^+$). Put $h = t(y - x)$. Since $y \neq x$, we have $h \neq 0$ for $t > 0$, and $h \to 0$ as $t \to 0^+$. Hence

$$\frac{g\big(x + t(y - x)\big) - g(x)}{t} = (y - x)\,\frac{g(x + h) - g(x)}{h} \;\longrightarrow\; (y - x)\,g'(x) \qquad (t \to 0^+),$$

by the definition of the derivative $g'(x)$, which exists because g is C^1 on the open interval I. The right-hand side of (1) does not depend on t, and limits preserve weak inequalities, so letting $t \to 0^+$ in (1) gives

$$g'(x)(y - x) \;\geq\; g(y) - g(x),$$

which is the required inequality $g(y) \leq g(x) + g'(x)(y - x)$. □

Remark. Geometrically, the graph of a concave g lies on or below each of its tangent lines. The proof only uses that g is differentiable at x.

Part (c)

Claim. Let $A \subseteq \mathbb{R}^n$ be open and convex and F a C^1 function on A. If F is concave on A, then

$$F(y) \le F(x) + \nabla F(x) \cdot (y - x) \qquad \text{for all } x, y \in A.$$

Proof. Fix $x, y \in A$ and let $v = y - x$. We reduce the claim to part (b) by restricting F to the part of the line through x and y that lies in A.

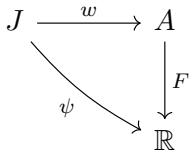
Step 1 (the one-variable function). Let

$$J = \{t \in \mathbb{R} : \ x + t(y - x) \in A\}, \qquad \psi : J \to \mathbb{R} \quad \psi(t) = F\big(x + t(y - x)\big).$$

By Lemma 1 (the general form of part (a)), applied with $v = y - x$, the set J is an open convex subset of $\mathbb{R}$, that is, an open interval, and ψ is concave on J. Moreover $0 \in J$ (because $x \in A$) and $1 \in J$ (because $x + 1 \cdot (y - x) = y \in A$). Note that $\psi(0) = F(x)$ and $\psi(1) = F(y)$.

Step 2 (ψ is C^1 and its derivative). We can represent the real valued map $\psi : J \to \mathbb{R}$ as two maps:

- Firstly, the map, $w : J \to A$, such that $w(t) = x + t(y - x)$ (which lies in A for $t \in J$, by the definition of J), and,
- Secondly, the map $F : A \to \mathbb{R}$.



The map w is differentiable from J into A; since w goes from $\mathbb{R}$ to $\mathbb{R}^n$, its Jacobian at t is the $n \times 1$ matrix (column vector) $Dw(t) = y - x$. Also F is continuously differentiable on A, with $1 \times n$ Jacobian $DF(u) = \nabla F(u)^\top$. By the chain rule $\psi = F \circ w$ is differentiable on J with $\psi'(t) = DF\big(w(t)\big)\, Dw(t)$, that is,

$$\psi'(t) = \sum_{i=1}^{n} \frac{\partial F}{\partial x_i}\big(w(t)\big)\,(y_i - x_i) = \nabla F\big(x + t(y - x)\big) \cdot (y - x).$$

Since each $\partial F / \partial x_i$ is continuous and w is continuous, ψ' is continuous. So ψ is a C^1 function on the open interval J. In particular

$$\psi'(0) = \nabla F(x) \cdot (y - x).$$

Step 3 (apply part (b)). ψ is a concave C^1 function on the open interval J, and $0, 1 \in J$. Part (b), applied to $g = \psi$ at the points 0 (in place of x) and 1 (in place of y), gives

$$\psi(1) \le \psi(0) + \psi'(0)\,(1 - 0).$$

Substituting the values from Steps 1 and 2,

$$F(y) \le F(x) + \nabla F(x) \cdot (y - x).$$

Since $x, y \in A$ were arbitrary, this proves the claim. □

Remark (A direct argument). One can also argue as in part (b) without passing through a one-variable function. For $t \in (0,1]$ the point $x + t(y-x) = ty + (1-t)x$ lies in A, and concavity gives

$$\frac{F\big(x + t(y-x)\big) - F(x)}{t} \;\geq\; F(y) - F(x).$$

Because F is C^1, it is differentiable at x, so the left-hand side tends to the directional derivative $\nabla F(x) \cdot (y-x)$ as $t \to 0^+$, and the inequality survives in the limit. The proof given above shows the structure of the test more clearly: (c) follows from (a) and (b).

Remark (Interpretation). The right-hand side $F(x) + \nabla F(x) \cdot (y-x)$ is the first-order (linear) Taylor approximation of F around x. The inequality says that, for a concave function, this linear approximation never underestimates the function: the graph of F lies on or below its tangent hyperplane at every point of A.

Part (d)

Claim. Let $A \subseteq \mathbb{R}^n$ be open and convex and F a C^2 function on A. If F is concave on A, then the Hessian matrix $H_F(x)$ is negative semidefinite for every $x \in A$, i.e.

$$v^\top H_F(x)\, v \leq 0 \qquad \text{for all } x \in A,\ v \in \mathbb{R}^n.$$

Proof. Fix $x \in A$ and $v \in \mathbb{R}^n$. If $v = 0$ then $v^\top H_F(x) v = 0$, so assume $v \neq 0$.

Step 1 (room to move). Since A is open, there is $r > 0$ such that the open ball $\{z : \|z - x\| < r\}$ is contained in A. Put $\delta = r/\|v\| > 0$. For $0 < t < \delta$ we have $\|(x + tv) - x\| = t\|v\| < r$, so $x + tv \in A$.

Step 2 (first-order upper bound from part (c)). Let $0 < t < \delta$. Since F is C^2 it is in particular C^1, and F is concave on the open convex set A. Part (c) with the points x and $x + tv$ gives

$$F(x + tv) - F(x) \;\leq\; \nabla F(x) \cdot (tv) = t\, \nabla F(x) \cdot v. \tag{2}$$

Step 3 (second-order expansion from Taylor's theorem). By Theorem 3 of the lecture notes, applied with $x^1 = x$ and $x^2 = x + tv$ (both in the open convex set A, with F twice continuously differentiable on A), there is $\theta(t) \in [0,1]$ (which in general depends on t, since the point $x^2 = x + tv$ does; for each t we fix one such value) such that

$$F(x + tv) - F(x) = t\, \nabla F(x) \cdot v + \frac{t^2}{2}\, v^\top H_F\big(z(t)\big)\, v, \tag{3}$$

where

$$z(t) = \theta(t)\, x + \big(1 - \theta(t)\big)(x + tv) = x + \big(1 - \theta(t)\big)\, t\, v.$$

Here we used $x^2 - x^1 = tv$, so $(x^2 - x^1)^\top H_F(z(t))(x^2 - x^1) = t^2\, v^\top H_F(z(t)) v$. Note that $z(t) \in A$ by convexity of A.

Step 4 (comparison). Substituting (3) into (2) and cancelling the common term $t\, \nabla F(x) \cdot v$ gives

$$\frac{t^2}{2}\, v^\top H_F\big(z(t)\big)\, v \;\leq\; 0, \qquad \text{hence} \qquad v^\top H_F\big(z(t)\big)\, v \;\leq\; 0 \quad \text{for every } t \in (0, \delta),$$

after dividing by $t^2/2 > 0$.

Step 5 (letting $t \to 0^+$). Since $0 \leq 1 - \theta(t) \leq 1$,

$$\|z(t) - x\| = \big(1 - \theta(t)\big)\, t\, \|v\| \;\leq\; t\, \|v\| \;\longrightarrow\; 0 \qquad (t \to 0^+),$$

so $z(t) \to x$ as $t \to 0^+$. The function $q : A \to \mathbb{R}$,

$$q(u) = v^\top H_F(u)\, v = \sum_{i=1}^{n} \sum_{j=1}^{n} v_i v_j \, \frac{\partial^2 F}{\partial x_i \partial x_j}(u),$$

is continuous on A, because F is C^2 and so all second-order partial derivatives are continuous. Therefore $q\big(z(t)\big) \to q(x)$ as $t \to 0^+$ (we do not need $t \mapsto \theta(t)$ to be continuous or even uniquely determined; only the bound $\|z(t) - x\| \le t\|v\|$ is used). Since $q\big(z(t)\big) \le 0$ for all $t \in (0, \delta)$, by Step 4, and limits preserve weak inequalities,

$$v^\top H_F(x)\, v = q(x) = \lim_{t \to 0^+} q\big(z(t)\big) \;\le\; 0.$$

As $x \in A$ and $v \in \mathbb{R}^n$ were arbitrary, $H_F(x)$ is negative semidefinite for every $x \in A$. □

Remark (An alternative route through one variable). One can also reduce (d) to a statement about functions of one variable, in the spirit of parts (a)–(c). Fix $x \in A$ and $v \in \mathbb{R}^n$, and let $J = \{t : x + tv \in A\}$ (an open interval containing 0) and $\psi(t) = F(x + tv)$. By Lemma 1, ψ is concave on J, and applying the chain rule twice gives $\psi \in C^2(J)$ with

$$\psi'(t) = \nabla F(x + tv) \cdot v, \qquad \psi''(t) = v^\top H_F(x + tv)\, v.$$

For $s < t$ in J, part (b) applied at s and at t gives

$$\psi(t) \le \psi(s) + \psi'(s)(t - s), \qquad \psi(s) \le \psi(t) + \psi'(t)(s - t).$$

Adding these yields $0 \le \big(\psi'(s) - \psi'(t)\big)(t - s)$. Since $t - s > 0$, this gives $\psi'(t) \le \psi'(s)$, so ψ' is nonincreasing on J. Consequently, for $h > 0$ small, $\big(\psi'(h) - \psi'(0)\big)/h \le 0$, and letting $h \to 0^+$ gives

$$v^\top H_F(x)\, v = \psi''(0) \le 0.$$

Remark (Why the conclusion is only "semidefinite"). Even strict concavity does not force the Hessian to be negative *definite*. For example, $F(x) = -x^4$ on $\mathbb{R}$ is strictly concave, but $F''(0) = 0$. So part (d) cannot be strengthened to negative definiteness.
